

Predictive Economic Connectedness among Guangdong Cities: An Exploratory PCA-VAR Network Audit

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Abstract

We revisit an archived PCA–VAR analysis of GDP for 21 Guangdong cities over 1999–2023. The retained outputs record 336 directed predictive links, corresponding to density 0.80 and average total degree 32, and the archived degree counts reconcile with those totals. Other outputs do not reconcile: a table labeled as Granger F statistics contains negative entries, and separate centrality graphics conflict with the recorded 336-edge graph. We therefore treat the network as an exploratory record of predictive connectedness rather than a calibrated significance or causal network. Recovery of the original data, code, and full test output is required for formal inference.

Keywords: economic geography; intercity connectedness; predictive networks; Guangdong

JEL Classification: C32; R11; R12; R15

1 Introduction

Cities are connected through trade, commuting, ownership ties, infrastructure, investment, and common exposure to macroeconomic conditions. Annual GDP does not observe those channels directly, but its lagged comovement can still reveal whether the past trajectory of one city contains predictive information about another. A directed network is a compact way to summarize that information, provided that its meaning is kept narrow.

That caution is important in economic geography. The network-city literature treats urban position as relational rather than as a simple function of population or output ([Batten, 1995](#); [Taylor, 2001](#); [Alderson and Beckfield, 2004](#)). Spatial econometrics formalizes another kind of dependence by starting from a spatial-weight structure ([Anselin, 1988](#)). Time-series predictive networks ask a different question again: whether lagged information improves

forecasts under a specified dynamic model. Granger’s original formulation is predictive, not a statement about structural intervention (Granger, 1969). Financial connectedness studies later showed how multivariate dynamics can be summarized as directed networks (Billio et al., 2012; Diebold and Yilmaz, 2014).

This paper examines an archived analysis of annual GDP for 21 Guangdong cities from 1999 to 2023. The original workflow used stationarity preprocessing, principal-component reduction, a VAR representation, pairwise predictive screening, and network summaries. During preparation of the present manuscript, however, the retained outputs were found not to be mutually consistent. One table labeled as Granger F statistics contains negative values, even though a conventional F statistic cannot be negative; other centrality and impulse-response graphics cannot be reconciled with the recorded 336-edge graph using their stated definitions. The raw city-year file, executable code, and complete testing output are no longer available in the retained project archive.

Rather than filling those gaps with reconstructed numbers, we take the inconsistency itself seriously. The analysis below reports only quantities that can be checked algebraically from the retained records: node count, edge count, density, and the degree profile encoded in the archived degree figure. The resulting object is an exploratory record of predictive connectedness. It is not a significance network, not a map of causal spillovers, and not a basis for ranking policy influence among cities.

This narrower treatment still serves an economic-geography purpose. It shows what can and cannot be inferred from a dense intercity predictive network when the provenance of the underlying statistical output is incomplete. It also sets out a reproducible extension protocol—including multiple-testing control and recovery of the original city-level series—that would be needed before substantive regional mechanisms could be evaluated.

2 Conceptual Positioning

2.1 Predictive networks versus spatial econometric dependence

Regional data can be related in more than one statistical sense. Classical spatial econometrics begins from a spatial-weight matrix and asks how outcomes or disturbances co-vary

across neighboring or otherwise connected locations (Anselin, 1988). A predictive network constructed from time-series information asks a different question: whether lagged information associated with one unit improves forecasts for another under a stated dynamic specification. The distinction matters because an edge in the present graph is not a spatial-lag coefficient and does not encode geographic contiguity. Two distant cities could be linked if their recorded dynamics are predictive, while adjacent cities need not be linked if the selected score falls below the threshold.

This separation also limits economic interpretation. Geographic proximity, industrial complementarity, common exposure to province-wide shocks, and sequential reporting or adjustment can all generate predictive dependence. A directed edge is therefore best viewed as a model-based empirical relation that can motivate later mechanism-specific analysis. Spatial econometric models, freight or commuting networks, and firm-level ownership links could subsequently test whether the same city pairs remain connected under alternative definitions of regional dependence. The present analysis does not substitute a predictive network for those approaches; it provides a different descriptive projection of the same regional system.

2.2 Dimension reduction, multiplicity, and network measurement

PCA is often used to summarize many correlated predictors by a smaller set of common components. In forecasting applications, principal-component methods can provide a parsimonious representation when the cross section is large relative to the available time dimension (Stock and Watson, 2002). That literature does not by itself validate the particular component count or preprocessing choices used here, because the retained materials do not preserve those tuning decisions. It does, however, clarify the statistical motivation for reducing dimension before dynamic modeling when only 25 annual observations are available.

A second issue arises after pairwise screening. With $21 \times 20 = 420$ ordered off-diagonal relations, a conventional hypothesis-testing analysis would create a substantial multiplicity problem. If valid p -values can be recovered, false-discovery-rate control such as the Benjamini–Hochberg procedure provides one principled way to distinguish a statistical

testing network from an exploratory threshold network (Benjamini and Hochberg, 1995). Because the present score matrix is signed and its original test definition is unavailable, we do not apply that procedure retrospectively to undocumented quantities. Instead, multiplicity is treated as a requirement for a future reproducible reanalysis.

Finally, graph statistics should be interpreted as distinct summaries rather than interchangeable measures of “importance.” Degree counts incident relations, whereas eigenvector-style centrality gives greater weight to connections involving already central nodes (Bonacich, 1972; Newman, 2010). Because the archived centrality output cannot be reconciled with the recorded 336-edge graph under the standard directed degree-centrality definition, the empirical discussion below uses degree counts only and does not rely on the composite or eigenvector-based ranking.

3 Data and Analytical Framework

3.1 Data structure and preprocessing

The analysis uses a compiled panel of annual GDP observations for 21 Guangdong cities covering 1999–2023, giving 25 annual observations per city. The retained methodological record states that the series were checked for stationarity and differenced when required before dynamic modeling. Principal-component analysis (PCA) was then used to reduce dimensionality before fitting a VAR. This reduction is practically motivated by the short time dimension: an unrestricted 21-variable VAR would contain many coefficients relative to 25 annual observations.

The retained dataset documentation does not include the original city-year data file, exact statistical-source links, or executable analysis code. Consequently, model settings not documented in the retained analytical record are not reconstructed. This limitation is material for replication and is addressed explicitly below.

3.2 PCA–VAR representation

Let Y_t denote the retained dynamic variables after preprocessing and PCA. A VAR of order p is

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t, \quad (1)$$

where c is an intercept, A_k are coefficient matrices, and u_t is an innovation vector. Standard VAR specification requires choices such as lag order, deterministic terms, and identification for impulse-response analysis (Lutkepohl, 2005). The retained analytical record does not preserve enough information to reproduce each of these choices exactly; results below are therefore restricted to quantities explicitly recorded in the analysis outputs.

3.3 Pairwise predictive scores and network construction

For an ordered city pair $i \rightarrow j$, Granger-style predictive screening asks whether lagged information associated with city i improves prediction of city j conditional on the specified model (Granger, 1969). The retained numerical output is a 21-by-21 matrix whose entries had been labeled as “Granger F statistics”. However, several entries are negative. A conventional F -statistic is nonnegative, so the signed matrix cannot be interpreted literally as a matrix of calibrated F tests.

To avoid turning a documentation inconsistency into a false statistical claim, we denote the retained entry for $i \rightarrow j$ by a generic score S_{ij} . The recorded graph construction retains an ordered pair when

$$S_{ij} > 2.5, \quad (2)$$

and uses the retained score as an edge weight. The threshold 2.5 is therefore treated only as an exploratory edge-retention rule. We do not attach a p -value to this cutoff, and we do not call a retained edge statistically significant without the missing degrees of freedom and test definition.

A fragment of the retained matrix is reproduced in Table 1. Its signed values are precisely why this cautious notation is necessary.

The network has one node per city. For a binary directed adjacency matrix $A = (a_{ij})$,

Table 1: Excerpt of the retained signed pairwise score matrix. Rows denote origin cities and columns denote destination cities.

	Dongguan	Zhongshan	Yunfu	Shanwei	Heyuan
Dongguan	0.000	5.413	-5.369	-5.421	-5.093
Zhongshan	4.195	0.000	3.956	2.043	2.946
Yunfu	-0.530	-0.412	0.000	-0.396	0.850
Shanwei	6.280	4.126	-3.826	0.000	3.140
Heyuan	3.052	0.850	-1.180	-1.698	0.000

in- and out-degree are

$$k_i^{\text{out}} = \sum_{j \neq i} a_{ij}, \quad k_i^{\text{in}} = \sum_{j \neq i} a_{ji}, \quad k_i = k_i^{\text{out}} + k_i^{\text{in}}. \quad (3)$$

For a directed graph with N nodes and E retained edges, density is $E/[N(N-1)]$. These identities allow the archived edge count and degree plot to be checked against one another without reconstructing the undocumented pairwise statistic.

4 Results

4.1 Aggregate network structure

Under the recorded threshold rule $S_{ij} > 2.5$, the analysis reports 336 directed edges among 21 cities. There are $21 \times 20 = 420$ possible off-diagonal ordered pairs, so the implied density is

$$\frac{336}{420} = 0.80. \quad (4)$$

The reported average total degree is also internally consistent with the edge count because $2E/N = 2 \times 336/21 = 32$. Table 2 summarizes these quantities.

Table 2: Recorded statistics for the thresholded directed network.	
Statistic	Recorded value
Nodes	21
Directed edges	336
Possible directed off-diagonal edges	420
Density	0.8000
Average total degree	32.00
Recorded total-degree range	26–38
Cities tied at total degree 38	Yunfu, Huizhou, Shanwei, Chaozhou

The main topological feature is therefore density rather than sparsity. Four fifths of all possible ordered links survive the chosen screening rule. A graph this dense can reflect genuine integration, common trends or shocks, or a permissive edge rule. It follows that any core–periphery interpretation is necessarily conditional on the threshold and on the undocumented construction statistic.

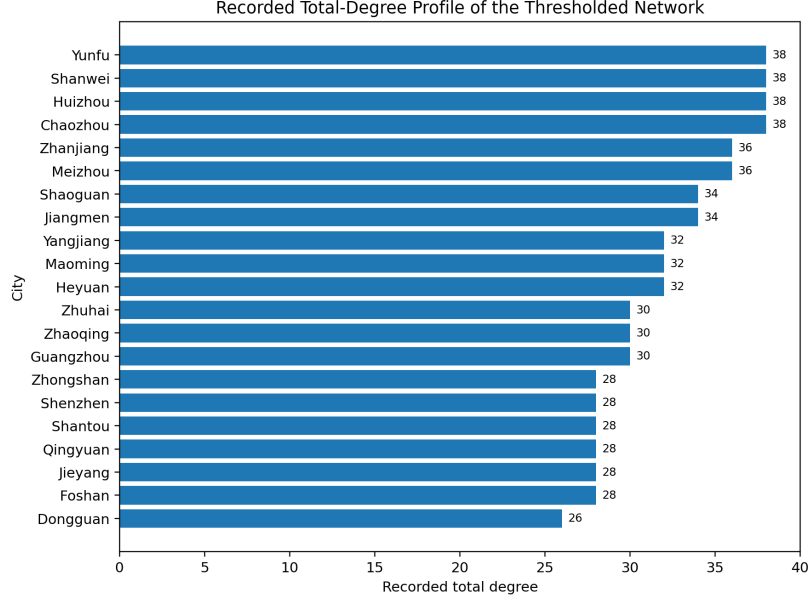


Figure 1: Recorded total-degree profile reconstructed from the archived degree-distribution figure. The original in-degree and out-degree bars each sum to 336; total degree therefore sums to 672 and averages 32 across 21 cities. Only total degree is shown because it is the component of the archived degree output that can be cross-checked directly against the reported edge count.

Alt text: Bar chart of total degree for 21 Guangdong cities in the archived threshold network. Values range from 26 to 38; Yunfu, Huizhou, Shanwei, and Chaozhou are tied at 38, while Dongguan is lowest at 26.

4.2 Recorded degree profile

Figure 1 shows the one archived node-level result that can be reconciled directly with the aggregate network statistics. Total degree ranges from 26 to 38. Yunfu, Huizhou, Shanwei, and Chaozhou are tied at 38 in the archived degree plot; there is therefore no unique highest-degree city. Dongguan has total degree 26 in that same plot. These values are

descriptive properties of the recorded threshold graph and should not be interpreted as a general ranking of regional economic importance.

The archived degree plot reports identical in-degree and out-degree counts for every city. This pattern is possible in a directed graph but unusually restrictive, and the missing code prevents us from determining whether it reflects the actual thresholded adjacency matrix or a plotting/construction choice. For that reason, the revised analysis does not infer distinct “influencer” and “receiver” roles from the degree output.

4.3 Output-consistency assessment

Several visually prominent archived outputs are excluded from the quantitative interpretation. The two heatmaps labeled as Granger F statistics are nonnegative, whereas the appendix table excerpt contains negative values under the same label. The centrality figures are also inconsistent with the 336-edge graph under the standard directed degree-centrality definition: the plotted degree-centrality values would correspond to a much sparser graph. The composite key-city ranking is derived from those same centrality values and is therefore not used. Finally, the impulse-response heatmap cannot be interpreted without the missing horizon, identification convention, and generating code. Excluding these plots avoids converting unresolved documentation differences into empirical claims.

4.4 Internal consistency checks

Although the original model cannot be fully rerun, several aggregate quantities can be checked algebraically. The 336 reported edges imply density $336/(21 \times 20) = 0.80$, exactly the recorded value. The same edge count implies an average total degree of $2E/N = 32$, again matching the record. These equalities show that the headline edge count, density, and average degree refer to the same directed graph rather than to different threshold settings.

The signed excerpt in Table 1 provides a different type of check: it reveals a labeling incompatibility rather than confirming one. Negative entries are incompatible with a conventional F statistic, which is a ratio of nonnegative quadratic forms. For that reason, the current paper deliberately retains the topology but not the original statistical label. This distinction is important for research integrity: an internally consistent graph can still

rest on a score whose inferential meaning is undocumented.

Table 3: What is directly supported by the retained record and what requires recovery of the original analysis.

Supported directly	Requires original data/code or test output
21 cities and 1999–2023 annual span	Exact city-year GDP source table and transformations
336 retained directed edges at the recorded cutoff	Formal F statistics, degrees of freedom, and p -values
Density 0.80 and average total degree 32	Number of retained principal components and lag-selection criterion
Archived degree counts summing to the reported edge total	Recomputed centrality, threshold stability, and multiple-testing correction
Signed table excerpt and separately archived nonnegative heatmaps	Exact pairwise statistic and mapping between archived outputs

5 Robustness and Interpretation

5.1 Threshold dependence

The cutoff of 2.5 is consequential. Because density is already 0.80, small threshold changes could materially alter degree rankings and the apparent core. A stronger analysis would rebuild the graph over a threshold grid and report density, strongly connected components, and rank correlations of centrality. If key-city rankings remain stable, the interpretation would become more persuasive; if rankings change sharply, node importance should be reported as threshold-dependent.

5.2 Multiple testing and sample length

There are 420 ordered off-diagonal pairs. Formal pairwise testing would therefore require reporting the underlying test statistic, degrees of freedom, and p -values, together with control of family-wise error or false discovery. None of these can be reconstructed from the signed score matrix alone. If the original analysis is recovered and produces valid pairwise p -values $p_{(1)} \leq \dots \leq p_{(m)}$, the Benjamini–Hochberg procedure would compare the ordered values with kq/m for a chosen false-discovery rate q and retain the largest admissible index k (Benjamini and Hochberg, 1995). Such a network would have a transparent inferential

meaning that the current cutoff does not possess.

The annual frequency is another constraint: 25 observations are few for rich multivariate dynamics after differencing and lagging. PCA reduces parameter dimension but does not create temporal information. Moreover, selecting too many components or lags can consume the limited degrees of freedom, whereas aggressive compression can remove city-specific variation that matters for pairwise prediction. A reproducible extension should therefore report the retained variance or another explicit component-selection criterion, information-criterion values for candidate VAR lag orders, and residual diagnostics. The purpose is not to maximize graph density, but to show that the network is stable to reasonable model choices.

5.3 A reproducible extension protocol

The limitations above suggest a concrete reanalysis rather than an unspecified call for “more robustness.” First, reconstruct the 21-city panel from documented statistical sources and archive the raw and transformed series. Second, state the stationarity decision for every series and the exact transformation applied. Third, choose the PCA dimension using a documented criterion and report component loadings sufficiently to understand which cities drive each factor. Fourth, select the VAR lag order from a prespecified candidate set and report the criterion used. Fifth, compute conventional pairwise predictive tests with their degrees of freedom and p -values. Sixth, apply an explicit multiplicity correction and compare the resulting inferential graph with the exploratory score-threshold graph. Seventh, repeat the network calculation across plausible thresholds, lag orders, and component counts, reporting rank correlations of centrality and changes in density.

A further extension would compare the predictive graph with an explicitly spatial benchmark. A contiguity or distance-based weight matrix could be used in a spatial econometric specification following the general framework of [Anselin \(1988\)](#). Agreement between spatial and predictive links would suggest a geographic component, while systematic discrepancies could identify long-distance economic connections or common-factor effects that deserve separate investigation. This comparison is proposed as future work; no spatial-lag estimates are claimed in the current results.

5.4 Regional interpretation

The high recorded density means that, under the original cutoff, most ordered city pairs were assigned an edge. That feature is descriptive before it is economic. Dense predictive networks can arise from common provincial trends, short time series, a permissive threshold, shared exposure to macroeconomic shocks, or genuine intercity transmission. With the retained archive we cannot separate those mechanisms.

The degree profile is therefore useful mainly as a map of how the archived thresholding procedure distributed links across cities. Yunfu, Huizhou, Shanwei, and Chaozhou each have recorded total degree 38, but that fact does not establish that they are the most economically influential cities in Guangdong. Such a claim would require a validated test statistic, a defensible treatment of multiplicity, and ideally mechanism-specific data such as trade, commuting, firm ownership, investment, or transport flows. The present network can suggest where a later investigation might look; it cannot supply that mechanism on its own.

6 Reproducibility Limitations

Three reproducibility limitations must be resolved before the analysis can support formal inferential claims. First, the exact city-year GDP source table and numerical data file are unavailable in the retained documentation. Second, the appendix table labeled as an F -statistic matrix contains negative entries, while the archived heatmaps under the same label use a nonnegative scale. Third, the centrality outputs are not reconcilable with the reported 336-edge graph under the standard directed degree-centrality definition. The executable code that would distinguish alternative constructions is unavailable.

These limitations are not repaired by relabeling. A fully reproducible revision should recover the original GDP panel, document the statistical source of every city-year observation, rerun stationarity and lag-selection diagnostics, reproduce the pairwise tests with nonnegative test statistics and associated p -values, apply a multiple-testing correction, rebuild the adjacency matrix, and recompute all node-level network statistics from that single verified graph. Until that is done, the contribution is exploratory network description rather than confirmatory regional econometrics.

7 Conclusion

The retained Guangdong project archive supports one firm descriptive result: the original thresholding procedure recorded a very dense 21-city directed network with 336 edges, density 0.80, and average total degree 32, and the archived in- and out-degree counts are consistent with those aggregates. It does not support the stronger claims that appeared in some earlier outputs. The negative entries in a table labeled as Granger F statistics, the conflicting centrality graphics, and the absence of the original data and code prevent a defensible reconstruction of significance, impulse responses, or structural causal transmission.

For that reason, the paper treats the network as an exploratory record of predictive connectedness. This choice reduces the number of substantive claims, but it preserves a clear boundary between what the archive shows and what would have to be re-estimated. A future version should recover the 21-city annual series, document their statistical sources, reproduce the preprocessing and VAR specification, report valid test statistics and p -values, control multiplicity across ordered city pairs, and then compare the resulting network with independent economic-geography channels. Until that work is done, the present evidence is descriptive rather than causal.

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Data availability

The retained materials include the numerical outputs and visual summaries reported here, but not the original 21-city annual GDP data file, exact source-table citations, or executable

analysis code. Those missing materials are therefore not claimed as publicly available. Complete independent replication requires recovery and documentation of the original panel and analysis code.

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